Name+Email ID

CSCI 5521: Machine Learning Fundamentals (Spring 2022)[1]

Midterm Exam

Due on Gradescope by 1pm, Mar 18

Instructions:

- This test has 4 questions, 100 points + 3 points (extra credits), on 4 pages.
- **For full credit, show how you arrive at your answers.**
- You have **24 hours** to complete and submit this test to gradescope.

1. **(32 points)** In I-IV, select the correct option(s) (it is not necessary to explain).

I. Select all the option(s) that correspond to supervised-learning algorithms:
(a) k-Means No Label
(b) Linear discriminant analysis yes label
(c) Linear discrimination yes label
(d) Linear regression continuous class label

II. What could we do to reduce overfitting? Select all the option(s) that apply:
(a) Add more training data and keep testing data the same
(b) Add more testing data and keep training data the same
(c) Change to a less complex model (e.g., a model with less parameters)
(d) Use mixture of Gaussians instead of k-means for clustering
more complicated, more parameter (high dimensional), extra info. more iteration need

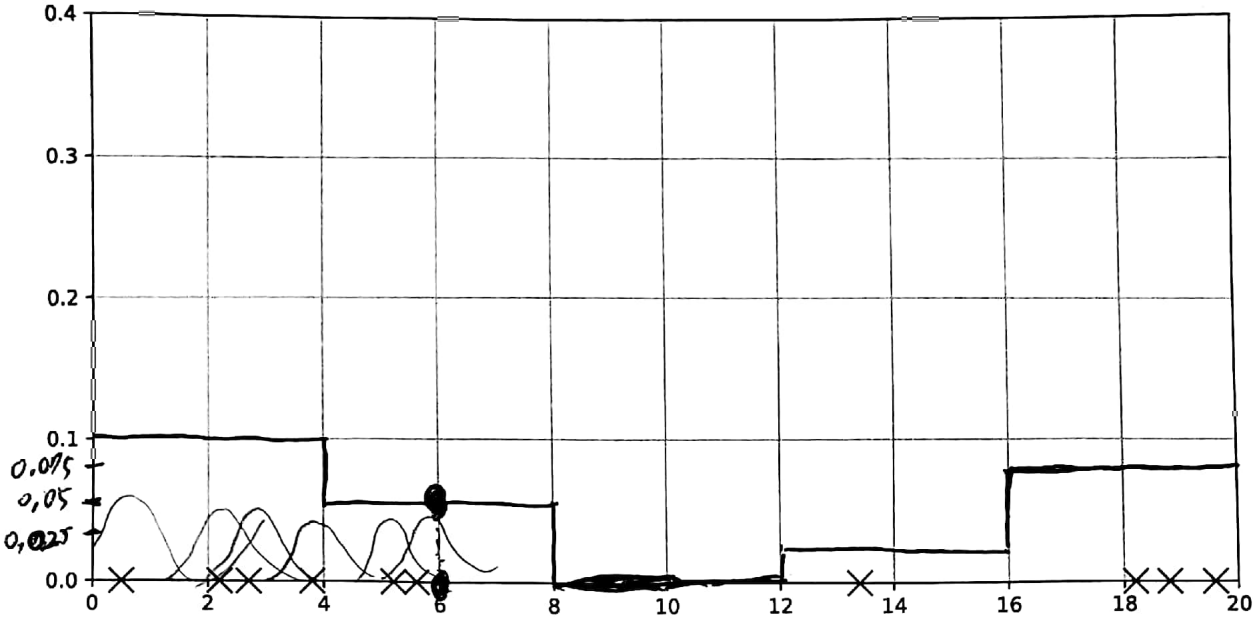
III. Select all the option(s) that correspond to dimensionality reduction algorithms:
(a) Linear discriminant analysis (b) Mixture of Gaussians
(c) Principal component analysis (d) Expectation maximization → k clustering.
→ ??

IV. Select all the true statement(s) below.:
(a) Both MLE and MAP are parametric. → both set parameter.
(b) Both MLE and MAP consider prior information. only MAP considers prior
(c) Expectation Maximization finds local optimum instead of global ones.
(d) Gradient descent finds global optimum instead of local ones.
Gradient descent → local.

[1]Instructor: Catherine Qi Zhao. TA: Prithvi Raj Botcha, Shi Chen, Suzie Hoops, James Yang, Yifeng Zhang. Email: csci5521.s2022@gmail.com

Scanned with CamScanner

3. **(24 points)** Answer the following questions about non-parametric density estimator:



(a) Draw a histogram estimator (start from origin) using $h = 4.0$ for the following 10 training data points in $\mathbb{R}$: 0.5, 2.2, 2.7, 3.8, 5.2, 5.6, 13.4, 18.2, 18.8, 19.6

(b) Given a test data point $x = 6.0$, what is the predicted density $p(x)$ for the data point? Show your work for full credits.

(c) List one possible approach to get a smoother density estimate. Draw an approximate curve when the kernel is used. You do not need to show the calculation.

Scanned with CamScanner

2.

(a) ◎ ⇒ class1 ⇒ only one data point in the neighborhood,
⇒ classify the label of the new data to the same class as its nearest neighbor,

(b) △ ⇒ class2 ⇒ For fixed $k=20$, using KNN estimator,

$$\hat{p}(C_1|x) = \frac{k_1}{k} = \frac{5}{20} = \frac{1}{4}$$

$$\hat{p}(C_2|x) = \frac{k_2}{k} = \frac{15}{20} = \frac{3}{4}$$

$$\hat{p}(C_2|x) > \hat{p}(C_1|x)$$

∴ new data is classified to class 2.

(c) ○ ⇒ class 1 ⇒ uniform kernel is pretty much similar as KNN, As long as the data is covered/counted in the neighborhood, in the range, in this case, d,

of ○ (class 1) = 28

of △ (class2) = 20

each of data has equal weight, and range covers every data (d) // and # of data in class 1 is bigger than the one in class 2,

∴ the label of new data point is class 1.

Scanned with CamScanner

3. (a) Histogram Estimator,

$$\hat{p}(x) = \frac{\#\{X^+ \text{ in the same bin as } x\}}{Nh}$$

i) for range (x_0, x_0+h):

$$\hat{p}(x) = \frac{4}{10.4} = \frac{1}{10} = 0.1$$

ii) for range (x_0+h, x_0+2h):

$$\hat{p}(x) = \frac{2}{10.4} = \frac{1}{20} = 0.05$$

iii) for range (x_0+2h, x_0+3h):

$$\hat{p}(x) = \frac{0}{10.4} = 0$$

iv) for range (x_0+3h, x_0+4h):

$$\hat{p}(x) = \frac{1}{10.4} = \frac{1}{40} = 0.025$$

v) for range (x_0+4h, x_0+5h):

$$\hat{p}(x) = \frac{3}{10.4} = \frac{3}{40} = 0.075$$

Scanned with CamScanner

(b)

given a test data. $x=6.0$, the test data is in range $(x-h, x+2h)$, which the pdf in the range from training data is 0.05

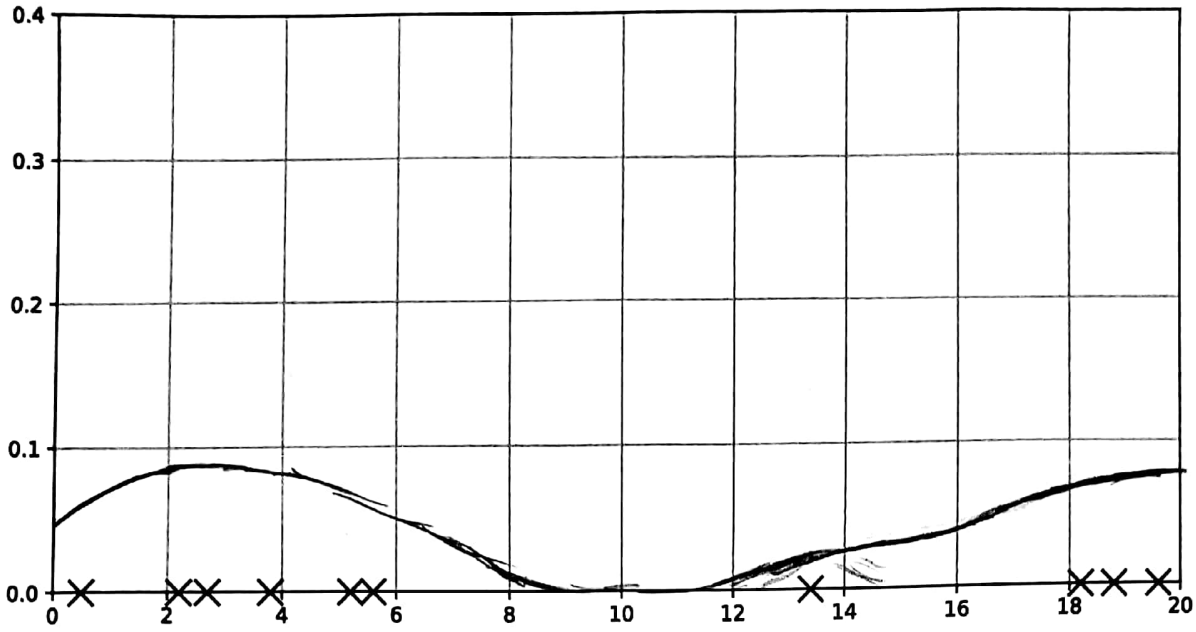
$$\therefore \hat{p}(6.0) = 0.05$$

(c) using kernel estimator with kernel function in Gaussian kernel, pdf can be smoother

By applying a kernel, the value of data point becomes the weighted average of data points nearby

Scanned with CamScanner

3. **(24 points)** Answer the following questions about non-parametric density estimator:



(a) Draw a histogram estimator (start from origin) using $h = 4.0$ for the following 10 training data points in $\mathbb{R}$: 0.5, 2.2, 2.7, 3.8, 5.2, 5.6, 13.4, 18.2, 18.8, 19.6

(b) Given a test data point $x = 6.0$, what is the predicted density $p(x)$ for the data point? Show your work for full credits.

(c) List one possible approach to get a smoother density estimate. Draw an approximate curve when the kernel is used. You do not need to show the calculation.

Scanned with CamScanner

4.

a) $L(\pi, \theta | X, z) = \log \prod_{n=1}^{N} \prod_{k=1}^{K} \left(\pi_k^{z_{nk}} \text{Bern}(x_n | \theta_k)^{z_{nk}} \right)$

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} \log \left(\pi_k^{z_{nk}} \left[\theta_k^{x_n} (1-\theta_k)^{1-x_n} \right]^{z_{nk}} \right)$$

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} \log \pi_k + \sum_{n=1}^{N} \sum_{k=1}^{K} x_n \cdot z_{nk} \log \theta_k + \sum_{n=1}^{N} \sum_{k=1}^{K} (1-x_n) \cdot z_{nk} \log (1-\theta_k)$$

↓ partial derivative : Lagrangian

$$\frac{\partial L(\pi, \theta | X, z)}{\partial \theta_k} = 0 + \frac{\sum_{n=1}^{N} \sum_{k=1}^{K} x_n \cdot z_{nk}}{\theta_k} - \frac{\sum_{n=1}^{N} \sum_{k=1}^{K} (1-x_n) z_{nk}}{1-\theta_k} + 0 = 0$$

$$\Leftrightarrow \frac{\sum_{n=1}^{N} \sum_{k=1}^{K} x_n \cdot z_{nk}}{\theta_k} = \frac{\sum_{n=1}^{N} \sum_{k=1}^{K} (1-x_n) z_{nk}}{1-\theta_k}$$

$$(1-\theta_k) \sum_{n=1}^{N} \sum_{k=1}^{K} x_n \cdot z_{nk} = \theta_k \cdot \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} - \theta_k \sum_{n=1}^{N} \sum_{k=1}^{K} x_n z_{nk}$$

$$\frac{\sum_{n=1}^{N} \sum_{k=1}^{K} x_n \cdot z_{nk}}{N} = \hat{\theta}_k$$

∴

Scanned with CamScanner

(b) Based on eq.,

$$L(\pi,\theta \mid X,Z)=\sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\log\pi_k+\sum_{n=1}^{N}\sum_{k=1}^{K} x_n\cdot z_{nk}\log\theta_k+\sum_{n=1}^{N}\sum_{k=1}^{K}(1-x_n)z_{nk}\cdot\log(1-\theta_k)$$

[illegible]

$$\frac{\partial L(\pi,\theta \mid X,Z)}{\partial \pi_k}=\frac{\partial\left[\sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\log\pi_k+\sum_{n=1}^{N}\sum_{k=1}^{K} x_n\cdot z_{nk}\log\theta_k+\sum_{n=1}^{N}\sum_{k=1}^{K}(1-x_n)z_{nk}\cdot\log(1-\theta_k)+\lambda\left(\sum_{k=1}^{K}\pi_k-1\right)\right]}{\partial \pi_k}=0$$

[illegible]

$$\Rightarrow \frac{\sum_{n=1}^{N} z_{nk}}{\pi_k}+\lambda=0$$

$$\Rightarrow \pi_k=-\frac{1}{\lambda}\sum_{n=1}^{N} z_{nk}$$

$$\Rightarrow \sum_{k=1}^{K}\pi_k=1=-\frac{1}{\lambda}\left(\sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}\right)$$

$$\left(\sum_{n=1}^{N}\sum_{k=1}^{K} z_{nk}=N\right)$$

$$\Rightarrow \lambda=-N$$

$$\Rightarrow \hat{\pi}_k=-\frac{\sum_{n=1}^{N} z_{nk}}{\lambda}=\frac{\sum_{n=1}^{N} z_{nk}}{N}$$

Scanned with CamScanner